

BARKER 2000

QUESTION 1. [12 marks]

- (a) If the surface area of a sphere is given by $A = 4\pi r^2$, find the radius (correct to 2 decimal places) if the surface area is 500 cm^2 .

[2m]

- (b) Find the value of e^2 correct to 3 significant figures.

[2m]

- (c) Rationalise the denominator of $\frac{2}{\sqrt{2} - 1}$

[2m]

- (d) Factorise fully $3 - 12x^2$

[2m]

- (e) Solve $x(x - 4) = 5$

[2m]

- (f) If $\sin \theta = \frac{2}{\sqrt{5}}$, evaluate $\tan \theta$ for $90^\circ < \theta < 180^\circ$

[2m]

QUESTION 2. [START A NEW PAGE] [12 marks]

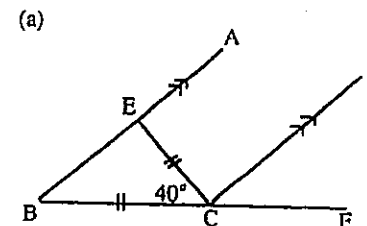


DIAGRAM NOT DRAWN TO SCALE

Given $AB \parallel CD$, $\angle BCE = 40^\circ$ and $BC = EC$,
find the size of $\angle ECD$, giving reasons.

[3m]

- (b) Draw a neat sketch showing the points $A(3, -1)$, $B(7, 2)$ and $C(1, 10)$ on a number plane.

- (i) Find the gradients of BC and of AB.

[2m]

- (ii) State why ABC is a right-angled triangle.

[1m]

- (iii) Find the area of $\triangle ABC$.

[2m]

- (iv) Find M, the midpoint of AC. Show that the equation of the line passing through M and parallel to the line AB is $3x - 4y + 12 = 0$.

[2m]

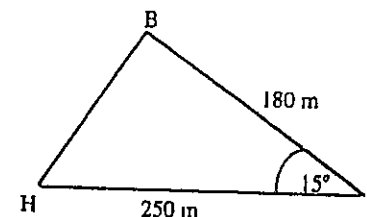
- (v) Find the equation of the circle which passes through the points A, B and C with centre M.

[2m]

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QUESTION 4. [START A NEW PAGE] [12 marks]

(a)



On a golf course, the distance from a tee, T, to the hole H is 250 metres. A golfer's ball comes to rest at point B, 180 metres from T. Angle HTB is 15° , as shown in the diagram. How far is B from H?

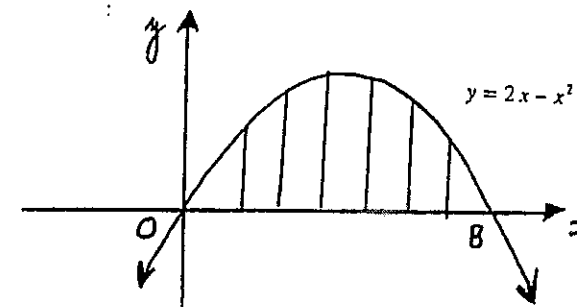
[3m]

(b) A supermarket displays cans of fruit in the form of a "pyramid". There are 3 cans in the top row, 5 in the next row, 7 in the next row and so on. If there are 20 rows on display, find:

- the number of cans in the bottom row;
- the total number of cans in the display.

[4m]

(c)



The diagram shows the graph of the function $y = 2x - x^2$.

- Find the x coordinate of the point B where the curve crosses the positive x -axis.
- Find the area of the shaded region contained by the curve $y = 2x - x^2$ and the x -axis.
- Write down a pair of inequalities that specify the shaded region.

[5m]

QUESTION 3. [START A NEW PAGE] [12 marks]

(a) Differentiate the following functions:

(i) $y = (4x + 3)^2$

[2m]

(ii) $y = xe^{4x}$

[2m]

(b) Find the exact value of $\int_0^2 e^{3x} dx$

[2m]

(c) (i) Sketch the graph of $y = |x - 2|$, showing all the main features.

[1m]

(ii) State the domain and range of $y = |x - 2|$

[1m]

(d) If $f(x) = \begin{cases} x^2 - 1 & \text{for } x > 1 \\ x & \text{for } x \leq 1 \end{cases}$

find the value of $f(-2) + f(1) + f(3)$

[2m]

(e) Evaluate $\lim_{x \rightarrow 0} \frac{2x}{x^2 + x}$

[2m]

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QUESTION 5. [START A NEW PAGE] [12 marks]

- (a) Consider the curve given by $f(x) = x^3 - 12x$
- Find $f'(x)$
 - Find the coordinates of the stationary points and determine their nature.
 - Find any points of inflexion.
 - Draw a neat sketch of the curve.
 - Show that $f(x)$ is odd.

[7m]

- (b) Given $\log_6 6 = 1.792$ and $\log_6 2 = 0.693$, evaluate $\log_6 24$

[2m]

- (c) Katrina's chance of obtaining a ticket to the Olympic Swimming Finals is 28%, but she has a 52% chance of obtaining a ticket to the Olympic Badminton Finals.

Using a tree diagram (or otherwise), find the probability that she obtains:

- both Final tickets;
- only one of the Finals Tickets.

[3m]

QUESTION 6. [START A NEW PAGE] [12 marks]

- (a) For the parabola $x^2 = 8(3 - y)$:
- Find the coordinates of the vertex.
 - Sketch this parabola on a number plane showing the vertex, focus and the equation of the directrix.
 - For what values of x is this parabola always positive?

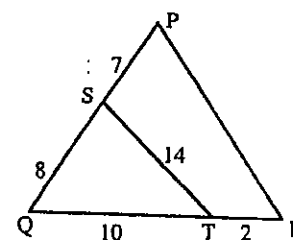
[6m]

- (b) Find the value of k for which the equation $3x^2 + 10x + k = 0$ has:

- one root which is the reciprocal of the other;
- equal roots.

[3m]

(c)

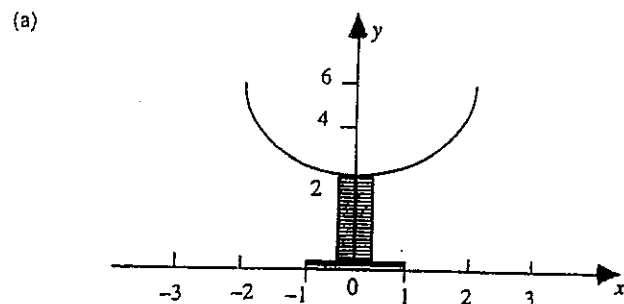
DIAGRAM IS NOT TO
SCALE

Given the diagram:

- Prove that $\triangle QST$ is similar to $\triangle QRP$.
- Hence find the length of PR .

[3m]

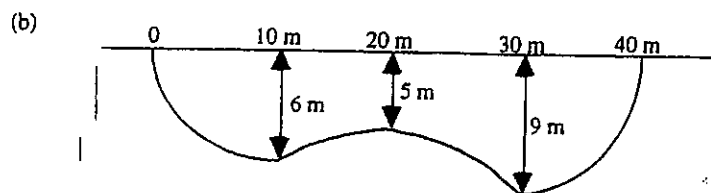
QUESTION 7. [START A NEW PAGE] [12 marks]



The bowl of a wine glass is formed by rotating the arc of the curve $y = x^2 + 2$ between (0, 2) and (2, 6) about the y-axis.

Find the volume of the bowl of the glass so formed.

[3m]



The diagram above represents a cross-section through a river, with the depth of the river marked every 10 metres as shown.

(i) COPY and COMPLETE this table in your paper.

x	0	10	20	30	40
$f(x)$					

(ii) Use Simpson's Rule with 5 function values to estimate the area of the cross-section.

(iii) If the river is flowing at the rate of 2 m per second, what volume of water passes through this cross-section each minute?

[5m]

(c) Find the equation of the curve $y = f(x)$, given that $\frac{d^2y}{dx^2} = 2x + 1$ and that there is a stationary point at (1, -2).

[4m]

QUESTION 8. [START A NEW PAGE] [12 marks]

(a) A plant is observed over a period of time. Its initial height is 30 cm. It grows 5 cm during the first week of observation. In each succeeding week the growth, in height, is 80% of the previous week's growth.

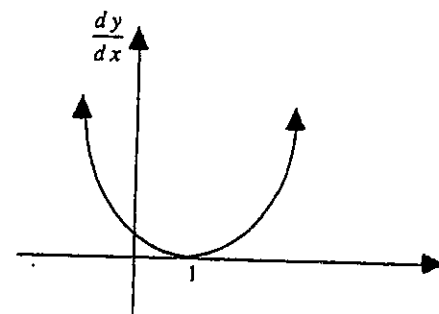
Assuming this pattern continues, calculate the plant's ultimate height.

[3m]

(b) Solve $4^x - 2(2^x) - 8 = 0$ for x

[4m]

(c) Consider the graph of the derivative $\frac{dy}{dx}$ given below.



(i) Comment on the sign of $\frac{dy}{dx}$ for all x except $x = 1$.

What does this imply about the curve $y = f(x)$ for all x , except $x = 1$?

(ii) What can you conclude about $y = f(x)$ when $x = 1$?

(iii) Sketch $\frac{d^2y}{dx^2}$

(iv) Sketch a possible graph of $y = f(x)$.

[5m]

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QUESTION 9. [START A NEW PAGE] [12 marks]

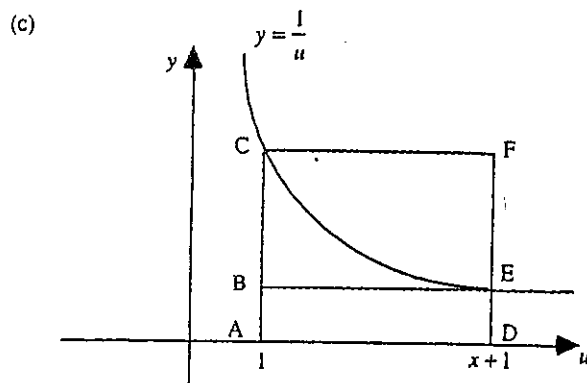
- (a) (i) Find the derivative of
- $y = e^{x^2}$

- (ii) Hence, evaluate
- $\int_0^2 x e^{x^2} dx$

[2m]

(b) $\int \frac{2x dx}{2 - 5x^2}$

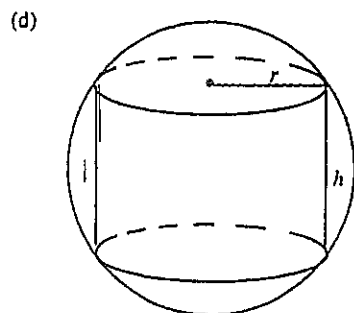
[2m]



Given this diagram and using the fact that area ABED < area ACED < area ACFD

show that $\frac{x}{x+1} < \ln(x+1) < x$

[3m]



A cylinder of radius (r) and height (h)
fits exactly inside a sphere of radius (R).

If the sphere has a radius of 12 cm:

- (i) show that the volume of the cylinder can be written as

$$V = \pi \left(144h - \frac{h^3}{4} \right)$$

- (ii) determine the exact value of the height (
- h
-) of the cylinder for it to be a maximum volume.

[5m]

QUESTION 10. [START A NEW PAGE] [12 marks]

- (a) Mr and Mrs Lee wish to buy a home unit in Hornsby. They obtained a loan of \$120 000 from a bank which they agreed to repay by equal monthly repayments. We assume that compound interest is calculated at a fixed rate of 7.2% p.a.

- (i) If the monthly repayment is \$ M , then show $A_1 = \$(120\,000 \times 1.006 - M)$ is the amount owing at the end of one month and hence write an expression for A_2 , the amount owing at the end of two months.

- (ii) Write an expression for A_n , the amount owing after n months.

- (iii) If the \$120 000 loan (including interest charges) is exactly repaid at the end of 25 years, write an expression for the monthly repayment, \$ M .

- (iv) Calculate the monthly repayment.

[4m]

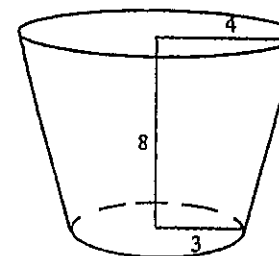
- (b) The curve $y = \frac{1}{2}(e^x + e^{-x})$ is called a catenary.

- (i) Draw a neat sketch of the catenary for $-2 \leq x \leq 2$.

- (ii) Hence, find the volume of the solid of revolution by rotating around the x -axis the region under the curve in (i) and above the x -axis and between $x = -2$ and $x = 2$.

[4m]

- (c) A drinking glass has the shape of a truncated cone.



- (i) If the internal radii of the base and the top are 3 cm and 4 cm respectively and the height is 8 cm, calculate the capacity of the glass.

- (ii) If the glass is filled with water to a depth of 4 cm, find the volume of the water.

[4m]

END OF EXAM

a) $A = 4\pi r^2$

$500 = 4\pi r^2$

$r = \sqrt{\frac{500}{4\pi}}$

≈ 6.31 (2 d.p.)

b) $e^2 \approx 7.39$

c) $\frac{2}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$

$= \frac{2\sqrt{2}+2}{2-1}$

$= 2\sqrt{2}+2$

d) $3(1-4x^2)$

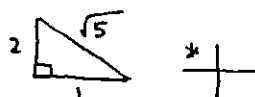
$= 3(1-2x)(1+2x)$

e) $x^2 - 4x - 5 = 0$

$(x-5)(x+1) = 0$

$x = 5, -1$

f)



$\therefore \tan \theta = -\frac{2}{1} = -2$

2.

a) $\triangle BEC$ is isos. (given)

$\therefore \angle CEB = 70^\circ$ ($\angle$ sum of isos. Δ)

$\therefore \angle ECD = 70^\circ$ (alt. $\angle$ s // lines)

b) i) BC: $m_1 = \frac{2-10}{7-1} = -\frac{4}{3}$

AB: $m_2 = \frac{2-1}{7-3} = \frac{3}{4}$

ii) $m_1 \times m_2 = -1$

iii) AB: $d = \sqrt{(7-3)^2 + (2-1)^2}$
 $= 5$ units

BC: $d = \sqrt{(7-1)^2 + (2-10)^2}$
 $= 10$ units.

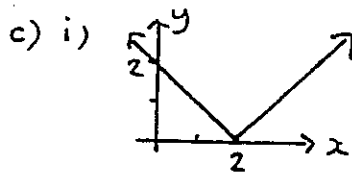
$A = \frac{1}{2} \times 10 \times 5$

Q3. a)

i) $y' = 2(4x+3) \times 4$
 $= 8(4x+3)$

ii) $y' = x \cdot 4e^{4x} + e^{4x}$
 $= e^{4x}(4x+1)$

b) $I = \frac{1}{3} [e^{3x}]_0^2$
 $= \frac{1}{3}(e^6 - e^0)$
 $= \frac{1}{3}(e^6 - 1)$



c) i) $x \in \mathbb{R}$
 $y \geq 0$

d) $-2 + 1 + (3^2 - 1)$
 $= 7$

e) Limit $= \lim_{x \rightarrow 0} \frac{2x}{x(x+1)}$
 $= \lim_{x \rightarrow 0} \frac{2}{x+1}$
 $= \frac{2}{0+1}$
 $= 2$

Q4. a)

$a^2 = b^2 + c^2 - 2bc \cos A$

$BH^2 = 180^2 + 250^2 - 2 \times 180 \times 250 \times \cos 15^\circ$

$BH^2 = 7966.675625$

$\therefore BH = 89.25623578 \text{ m}$
 ≈ 89.3 (1 d.p.)

b) 3, 5, 7, ...

AP. $a = 3, d = 2$

i) $T_n = a + (n-1)d$

$T_{20} = 3 + 19 \times 2$
 $= 41$

$\therefore 41$ cans in bottom row

iv) $M = (2, 4\frac{1}{2})$ $M_{AB} = \frac{3}{4}$

Q4. b) ii) $S_{20} = \frac{20}{2}(3+41)$
 $= 440$

440 cans in display.

c) i) $y = 2x - x^2$
 $y = x(2-x)$

let $y = 0 \therefore x(2-x) = 0$
 $x = 0$ or 2
 $\therefore B(2, 0)$

ii) $A = \int_0^2 (2x - x^2) dx$
 $= [x^2 - \frac{x^3}{3}]_0^2$
 $= (4 - \frac{8}{3}) - 0$
 $= \frac{4}{3} \text{ units}^2$

iii) $y \geq 0, y \leq 2x - x^2$

Q5. a) $f(x) = x^3 - 12x$

i) $f'(x) = 3x^2 - 12$

ii) let $f'(x) = 0$

$\therefore 3x^2 - 12 = 0$

$3(x^2 - 4) = 0$

$x = +2, -2$

when $x = 2, f(2) = -16 \therefore (2, -16)$

when $x = -2, f(-2) = 16 \therefore (-2, 16)$

$f''(x) = 6x$

when $x = 2, f''(2) = 12 \therefore$ min

when $x = -2, f''(-2) = -12 \therefore$ max

min $(2, -16)$ max $(-2, 16)$

iii) $f''(x) = 6x$

let $f''(x) = 0$

$\therefore 6x = 0$

$x = 0, y = 0$

x	0^-	0	0^+
$f''(x)$	$-$	0	$+$

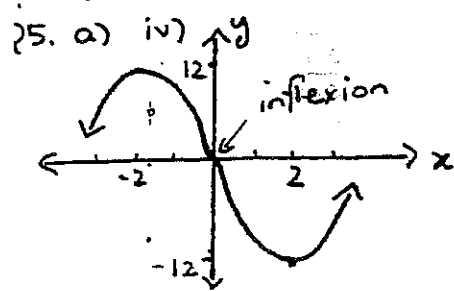
$f''(x)$ change sign

$\therefore$ point of inflexion at $(0, 0)$

$4y - 18 = 3x - 6$

$\therefore 3x - 4y + 12 = 0$

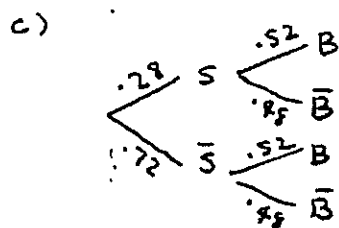
$\therefore$ diameter $= \sqrt{125} = 5\sqrt{5}$
radius $= \frac{5}{2}\sqrt{5}$, centre



$$\begin{aligned} \text{v) } f(-x) &= (-x)^3 - 12(-x) \\ &= -x^3 + 12x \\ &= -(x^3 - 12x) \\ &= -f(x). \end{aligned}$$

since $f(-x) = -f(x)$
 $f(x)$ is odd.

$$\begin{aligned} \text{b) } \log_a 24 &= \log_a (6 \times 4) \\ &= \log_a (6 \times 2^2) \\ &= \log_a 6 + 2 \log_a 2 \\ &= 1.792 + 2 \times 0.693 \\ &= 3.178 \end{aligned}$$



$$\begin{aligned} \text{i) } P(SB) &= 0.28 \times 0.52 \\ &= 0.1456 \end{aligned}$$

$$\begin{aligned} \text{ii) } P(S\bar{B}) + P(BS) &= 0.28 \times 0.48 + 0.52 \times 0.72 \\ &= 0.5088. \end{aligned}$$

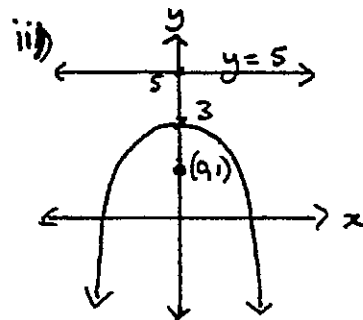
$$\begin{aligned} \text{5. a) i) } x^2 &= 8(3-y) \\ x^2 &= -8(y-3) \\ (x-h)^2 &= -4a(y-k) \end{aligned}$$

i) vertex $(0, 3)$

i) focal length $4a = 8$
 $\therefore a = 2$

foci $(0, 1)$

directrix $y = 5$.



$$\text{iii) let } y = 0$$

$$\therefore x^2 = 24$$

$$x = \pm 2\sqrt{6}.$$

Parabola is +ve

$$-2\sqrt{6} < x < 2\sqrt{6}.$$

b) i)

let roots be $\alpha, 1/\alpha$.

$$\text{prod. of roots} = \frac{c}{a} = 1.$$

$$\text{for } 3x^2 + 10x + k = 0$$

$$\alpha \cdot \frac{1}{\alpha} = \frac{k}{3}$$

$$1 = \frac{k}{3}$$

$$\therefore k = 3.$$

$$\text{ii) } \Delta = 0$$

$$b^2 - 4ac = 100 - 12k = 0$$

$$k = \frac{100}{12}$$

$$\therefore k = \frac{25}{3}.$$

$$\text{c) i) } \frac{QS}{QR} = \frac{QT}{QP}$$

$$\frac{8}{12} = \frac{10}{15}$$

$$\frac{2}{3} = \frac{2}{3}.$$

$$\angle SQT = \angle PQR$$

$$\therefore \Delta QST \sim \Delta QRP.$$

2 sides in proportion
 and 1 angle (inc.) equal.

$$\text{ii) } \frac{ST}{PR} = \frac{2}{3}$$

$$\frac{14}{PR} = \frac{2}{3}$$

$$\begin{aligned} \text{Q7. a) } V &= \pi \int_2^6 x^2 \cdot dy \\ &= \pi \int_2^6 y-2 \cdot dy \\ &= \pi \left[\frac{y^2}{2} - 2y \right]_2^6 \\ &= \pi (18 - 12 - (2 - 4)) \\ &= 8\pi \text{ units}^3. \end{aligned}$$

$$\text{b) i) } x \quad 0 \quad 10 \quad 20 \quad 30 \quad 40$$

$$f(x) \quad 0 \quad 6 \quad 5 \quad 9 \quad 0$$

$$\text{ii) } A = \frac{h}{3} \{ 0 + 0 + 4(6+9) + 2 \times 5 \}$$

$$= 233 \frac{1}{3} \text{ m}^2 \quad (h=10)$$

$$\begin{aligned} \text{iii) } V &= A \times H \\ &= 233 \frac{1}{3} \times 120 \quad \left(\begin{array}{l} 2\text{m/sec} \\ = 120\text{m} \end{array} \right) \\ &= 28000 \text{ m}^3 \end{aligned}$$

$$\text{c) } \frac{dy}{dx} = x^2 + x + c.$$

$$\text{when } x=1, \frac{dy}{dx} = 0.$$

$$\therefore 1^2 + 1 + c = 0 \Rightarrow c = -2$$

$$\therefore \frac{dy}{dx} = x^2 + x - 2.$$

$$y = \frac{x^3}{3} + \frac{x^2}{2} - 2x + k \quad \left(\begin{array}{l} \text{C} \\ \text{I} \end{array} \right)$$

$$\text{when } x=1, y=-2.$$

$$\therefore -2 = \frac{1}{3} + \frac{1}{2} - 2 + k$$

$$\therefore k = -5/6$$

$$\therefore y = \frac{x^3}{3} + \frac{x^2}{2} - 2x - \frac{5}{6}.$$

$$\text{Q8. a) } a=5, r=0.8$$

$$S_{\infty} = \frac{a}{1-r} = \frac{5}{1-0.8} = 25.$$

$$\therefore \text{Ult. height} = 30 + 25 = 55\text{cm}$$

$$\text{b) } 4^x - 2(2^x) - 8 = 0.$$

$$\text{let } m = 2^x$$

$$m^2 - 2m - 8 = 0$$

$$(m-4)(m+2) = 0$$

$$\therefore m = 4, -2$$

$$\therefore 2^x = 4 \quad \text{or} \quad 2^x = -2.$$

$$x = 2$$

$$x = -1$$

$$\frac{dy}{dx} > 0 \text{ for all } x \neq 1.$$

The function has positive gradient, except when $x=1$.

There is a stationary point on $y=f(x)$ as $\frac{dy}{dx}=0$ and if

$x < 1$, $\frac{dy}{dx} > 0$.

$x > 1$, $\frac{dy}{dx} > 0$.

horizontal point of inflexion at $x=1$.

Sketch of $y=f(x)$ as $x \rightarrow 0$ and if

$x < 1$, $\frac{dy}{dx} > 0$.

$x > 1$, $\frac{dy}{dx} > 0$.

horizontal point of inflexion at $x=1$.

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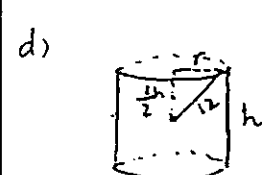
horizontal point of inflexion at $x=1$.

$$\text{Area } ACFD = [(x+1)-1] \times \frac{1}{x+1}$$

$$= \frac{x}{x+1}$$

$$\therefore \frac{x}{x+1} < \ln(x+1) < x$$

d)



$$r^2 = 12^2 - \left(\frac{h}{2}\right)^2$$

$$r^2 = 144 - \frac{h^2}{4}$$

$$\begin{aligned} \text{i) } V &= \pi r^2 h \\ &= \pi \left(144 - \frac{h^2}{4}\right) h \\ &= \pi \left(144h - \frac{h^3}{4}\right) \end{aligned}$$

$$\text{ii) } \frac{dV}{dh} = \pi \left(144 - \frac{3h^2}{4}\right)$$

$$\text{let } \frac{dV}{dh} = 0$$

$$\therefore \frac{3h^2}{4} = 144$$

$$h^2 = 192$$

$$h = \pm \sqrt{192}$$

but h is height

$$\therefore h = \sqrt{192}$$

h	10	$\sqrt{192}$	20
$\frac{dV}{dh}$	+69	0	-156
	+	0	-

$$\therefore \text{maximum volume when } h = \sqrt{192} \text{ cm}$$

$$\therefore h = 8\sqrt{3} \text{ cm}$$

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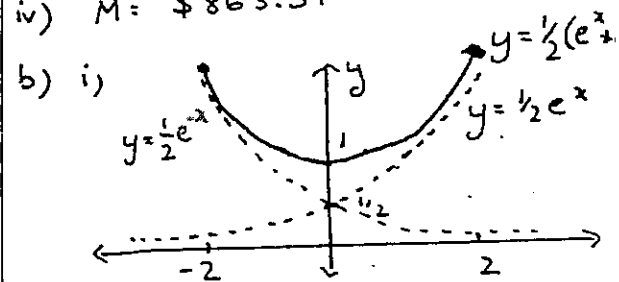
$$\therefore h = 8\sqrt{3} \text{ cm}$$

$$\therefore 0 = 120000 \times 1.006^{300} - M(1 + 1.006 + \dots + 1.006^{299})$$

$$0 = 120000 \times 1.006^{300} - M \left[\frac{1(1.006^{300} - 1)}{0.006} \right]$$

$$\therefore M = 120000 \times 1.006^{300} \times \frac{0.006}{1.006^{300} - 1}$$

$$\text{iv) } M = \$863.51$$



$$\text{ii) } V = \pi \int_{-2}^2 \left[\frac{1}{2}(e^x + e^{-x}) \right]^2 dx$$

$$= \frac{2\pi}{4} \int_{-2}^2 (e^{2x} + 2 + e^{-2x}) dx$$

$$= \frac{\pi}{2} \left[\frac{e^{2x}}{2} + 2x - \frac{e^{-2x}}{2} \right]_{-2}^2$$

$$= \frac{\pi}{2} \left[\left(\frac{e^4}{2} + 4 - \frac{e^{-4}}{2} \right) - \left(\frac{e^0}{2} + 0 - \frac{e^0}{2} \right) \right]$$

$$= \frac{\pi}{2} \left[\left(\frac{e^4}{2} + 4 - \frac{e^{-4}}{2} \right) - \left(\frac{1}{2} + 0 - \frac{1}{2} \right) \right]$$

$$= \frac{\pi}{2} (e^4 + 8 - e^{-4}) \text{ units}^3$$

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$$\text{c) i) } \begin{array}{c} 4 \\ 8 \\ x \end{array} \quad \begin{array}{c} \text{similar } \Delta s \\ \therefore \frac{x}{x+8} = \frac{3}{4} \\ 4x = 3x + 24 \\ \therefore x = 24 \end{array}$$

$$\text{Vol} = \text{vol. of big cone} - \text{vol. small cone}$$

$$= \frac{1}{3} \pi \cdot 4^2 \cdot 32 - \frac{1}{3} \pi \cdot 3^2 \cdot 24$$

$$= 98 \frac{2}{3} \pi \text{ cm}^3$$

$$\text{OR } \begin{array}{c} y \\ (1,8) \\ (3,0) \\ -24 \end{array} \quad \begin{array}{c} m = \frac{8-0}{4-3} = 8 \\ \text{equation of line:} \\ y-0 = 8(x-3) \\ y = 8x-24 \end{array}$$

$$\therefore y \text{ intercept} = -24$$

$$\text{if } y = 8x - 24 \Rightarrow x = \frac{y+24}{8}$$

$$V = \pi \int_0^8 x^2 dy = \pi \int_0^8 \left(\frac{y+24}{8} \right)^2 dy$$

$$= 98 \frac{2}{3} \pi \text{ units}^3$$

$$\text{ii) } V = \pi \int_0^4 \left(\frac{y+24}{8} \right)^2 dy$$

$$= \frac{127}{3} \pi \text{ cm}^3$$

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